

LIGHT EMITTING DIODES

Next-Generation Smart Lighting Sources

As energy efficiency becomes increasingly important for controlling costs, improving energy independence, and reducing environmental impacts, governments and organisations are looking up to LEDs as a solution



■ DR S.S. VERMA

Light-emitting diodes were invented in the 1920s. Today, LEDs come in various colours, brightness and wavelength—visible, ultraviolet and infrared. LEDs have a lot of advantages over traditional lighting including low energy consumption, long lifetime, small size and robustness. Everyone is looking at the next wave of lighting luminaires that will produce smart lighting with LED and organic light emitting diode (OLED) systems.

LEDs are more rugged, resembling something closer to hard plastic, than thin glass. While fluorescent lights have been justly praised for offering greater efficiency than incandescent bulbs, LEDs are more efficient and last longer than any other lighting source in the market, making them the next logical step. They are also environmentally sound, since their manufacturing does not require toxic substances such

as mercury.

Consumers are already getting their first taste of LED technology with indoor/outdoor festival lights. Experts expect that a new wave of lighting devices based on LEDs and solid-state lighting will supplant the common light bulb in coming years, leading to vast environmental, energy and cost benefits, as well as innovations in healthcare, transport systems, digital displays and computer networking.

A new report finds that the market for LEDs is set to explode in the next ten years, surpassing even compact fluorescent light bulbs as the ultimate energy-efficient lighting option. According to a study, by 2020, LEDs will cover 46 per cent of the \$4.4 billion US market for lamps in the commercial, industrial and outdoor stationary sectors, driven by the overall push toward energy efficiency. That's a big jump, considering that LEDs now capture only two per cent of the lighting market, according to estimates.

Advantages

LED bulbs typically last 50,000 hours—twice as long as fluorescent—and are up to twice as efficient as fluorescent, which are again four to six times more efficient than incandescent. Further, unlike fluorescent or incandescent bulbs, LED tubes do not get very hot. According to a report, incandescent bulbs only convert about four to six per cent of the energy they receive into light; the rest is wasted as heat. While fluorescent bulbs can raise the temperature in a room by up to two degrees resulting in increased cooling load for buildings, LEDs don't raise the temperature at all.

However, smart LEDs should not be seen as just a replacement technology for compact fluorescent lamps (CFLs) and incandescent lamps, but as a serious disruptive technology. Regular bulbs can only be turned on or off, whereas LEDs can be tuned like a radio.

LEDs have novel capabilities that make them extremely useful when going beyond the replacement paradigm. As an alternative to the traditional incandescent light bulb, LED lights provide significant energy-savings. They can be 2000 per cent more efficient than conventional light bulbs and 500 per cent more efficient than compact fluorescent bulbs.

Engineers and scientists predict that widespread use of LEDs over the course of ten years would save more than \$1000 billion in energy costs, eliminate the need for nearly a billion barrels of oil over ten years, and lead to a substantial reduction in emissions of carbon dioxide—the most common greenhouse gas.